



Yamaha YRM20 (Credit: <https://global.yamaha-motor.com>)

The larger components may require larger feeders. So, it is advisable to figure out the number of slots needed for larger tape feeder types, say, from 4mm to 56mm.

Mycronic MY300 has an 'intelligent feeder' solution that is claimed to take less than ten seconds of loading time. Agilis tape and stick feeders and tray handling systems in the machine are said to handle a wide range of components to meet the needs of today's SMT manufacturing grounds. The compact 4, 8, 12, and 16mm Agilis tape feeders come with an innovative cover-tape separator on the pen-sized feeder that exposes components for easy picking. The feeders provide quantity and location tracking, operator guidance, simplified data entry, moisture sensitive device (MSD) handling, and on-the-fly changeovers. Fully operator-independent, it ensures higher productivity in a smaller footprint. Its product set-up time is another important feature for quick processing.

With the SMT measuring feeder feature in place, there is no need to measure materials manually, which improves the efficiency and avoids

the occurrence of material error. The recent machines feature component verification before placement to ensure the right component is placed at the right location and comes with RCDT (resistor, capacitor, diode, and transistor) measurement tools. Neoden 8 features pneumatic feeders.

Component lead pitch. Industry experts consider 0.012 fine pitch as an unsaid thumb rule for selecting PnP machines. The component lead pitch is the distance

between the centres of the leads. However, experts warn buyers to not be moved by motor specifications or accuracy as even machines with great motor accuracy may not pick up, inspect, and place fine-pitch components.

Smart bin system. The most common reason for unplanned downtime in high-mix electronics manufacturing is missing material. A smart bin system helps machine operators minimise downtime by ensuring that all needed material is collected and prepared just-in-

time for production. Building on other powerful material handling solutions, the system brings ease-of-use to the kitting and changeover process. The end result is faster changeovers and higher utilisation.

Next-generation precision. The latest pick-and-place solutions combine rigid machine frame, advanced mount head technology, and automatic thermal adaption for the highest levels of precision. The MY300 adds an improved line scan vision system to take this accuracy into the next generation. It combines three programmable light sources with either 2K or 4K resolution to ensure a future-proof solution for the most advanced components down to 0.15mm pitch. Both options provide extremely precise on-the-fly positioning and inspection for any type of miniaturised or specialised component

Accuracy and repeatability. Experts' candid recommendation regarding accuracy for PnP machines is 0.00254mm (0.0001-inch) with a fine pitch lead of 12mil repeatedly. Though such high accuracy is commonly characterised for high-budget machines, one can always find a used PnP machine to attain such results. Besides, remember to buy a machine that comes with software suites to handle the process.

Featuring a single turret head



Eight synchronised nozzle heads of Neoden 8